Saarthi Manual

Section 3: Wiring Diagram and Circuit Setup

❓ 3.1 OVERVIEW: WHAT THIS SECTION COVERS

By the end of this section, you will be able to:
✅ Connect all electronic components correctly
✅ Understand which pin goes where and why
✅ Avoid power mistakes that could damage the system
✅ Create a neat, safe, and modular wiring setup
✅ Test basic component connectivity

🔧 3.2 MAIN COMPONENTS INVOLVED

Component	Description
Raspberry Pi 5	Main processor (You already have it)
Google Coral USB Accelerator	Speeds up AI model inference
Ultrasonic Sensors (x2 or x3)	Detect obstacles in front and downward
Raspberry Pi Camera (Night Vision optional)	Captures real-time video feed
Power Bank (20,000 mAh+)	Powers the Pi and Coral USB
Speaker or Earphones	Outputs audio alerts
Jumper Wires (Male-Female)	For GPIO and power connections
Waterproof Enclosure Box	Protects electronics (placed in pouch)
Soldering Kit (optional)	For strong, permanent connections

🧠 3.3 WIRING FLOWCHART (BIRD'S EYE VIEW)

```
css
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[Ultrasonic Sensors]  -->  [GPIO Pins of Raspberry Pi]
[Camera]              -->  [CSI Port of Pi]
[Speaker]             -->  [3.5mm Audio Jack or USB Audio Adapter]
[Google Coral USB]    -->  [USB 3.0 Port on Pi]
[Power Bank]          -->  [USB-C Port on Pi + USB-A for Coral]
```

Now, let’s break each of these down.

🔌 3.4 STEP-BY-STEP WIRING GUIDE

□ A. CONNECTING ULTRASONIC SENSOR (HC-SR04)

📌 We'll use 2 ultrasonic sensors:

- One facing front
- One angled downward for stairs

Each HC-SR04 has 4 pins:

1. **VCC** – Power (connects to 5V)
2. **Trig** – Trigger pin (GPIO OUT)
3. **Echo** – Echo pin (GPIO IN)
4. **GND** – Ground

□ Example Wiring (Front Sensor):

Sensor Pin	Raspberry Pi Pin
VCC	Pin 2 (5V)
GND	Pin 6 (GND)
Trig	GPIO 23 (Pin 16)
Echo	GPIO 24 (Pin 18)

□ Example Wiring (Downward Sensor):

Sensor Pin	Raspberry Pi Pin
VCC	Pin 4 (5V)
GND	Pin 9 (GND)
Trig	GPIO 17 (Pin 11)
Echo	GPIO 27 (Pin 13)

⚠□ IMPORTANT TIP:
Use **resistor voltage divider** on the **Echo** pin to bring the 5V signal down to 3.3V for the Pi (Echo → 1kΩ and 2kΩ resistors).

□ B. CONNECTING THE CAMERA (Raspberry Pi Camera Module)

📌 Steps:

1. Locate the **CSI (Camera Serial Interface)** port on your Pi.
2. Gently lift the plastic clip.
3. Insert camera ribbon cable with metal connectors facing the **HDMI port side**.
4. Push clip down to lock.

✅ Use command `raspi-config` → `Interface Options` → Enable camera
✅ Test with `libcamera-still -o test.jpg`

□ C. CONNECTING THE GOOGLE CORAL USB ACCELERATOR

📌 **Steps:**

1. Plug Coral USB Accelerator into one of the **blue USB 3.0 ports** on your Pi 5.
2. Make sure it's snug and secured.

⚠□ Needs **5V/900mA or more**, so ensure your power bank can supply enough current via USB-A.

□ D. CONNECTING THE SPEAKER (or EARPHONES)

You have **two choices**:

1. 3.5mm Speaker (Analog)

- Plug into Raspberry Pi's headphone jack (use `alsamixer` to control volume)

2. USB Speaker / USB Audio Adapter

- Plug into USB 2.0 port
- Use command: `aplay -l` to detect
- Set default device using `alsamixer`

□ E. POWERING THE PI AND CORAL USB

🔌 **With Power Bank:**

Component	Connection
Raspberry Pi	USB-C output of power bank (5V 3A recommended)
Coral USB	USB-A output of same power bank or via Pi USB port

⚠□ Use high-quality cables or you'll face power drops.

□ F. WIRING DIAGRAM (TEXT DESCRIPTION)

If we describe the wiring layout in one compact visual:

```
[Ultrasonic Sensor - Front]
- VCC → Pi 5V
- GND → Pi GND
- Trig → GPIO 23
- Echo → GPIO 24 (via 2kΩ + 1kΩ voltage divider)
```

```
[Ultrasonic Sensor - Downward]
 - Trig → GPIO 17
 - Echo → GPIO 27 (with voltage divider)

[Camera Module]
 - Connected to CSI port

[Google Coral USB]
 - USB 3.0 (Blue port)

[Speaker]
 - 3.5mm jack or USB audio

[Power]
 - Power bank USB-C → Pi
 - Power bank USB-A → Coral USB (optional)
```

□ 3.5 BASIC HARDWARE TESTING

✅ Ultrasonic Sensor Test (Python):

```
python
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import RPi.GPIO as GPIO
import time

TRIG = 23
ECHO = 24

GPIO.setmode(GPIO.BCM)
GPIO.setup(TRIG, GPIO.OUT)
GPIO.setup(ECHO, GPIO.IN)

GPIO.output(TRIG, False)
time.sleep(2)

GPIO.output(TRIG, True)
time.sleep(0.00001)
GPIO.output(TRIG, False)

while GPIO.input(ECHO)==0:
    pulse_start = time.time()

while GPIO.input(ECHO)==1:
    pulse_end = time.time()

pulse_duration = pulse_end - pulse_start
distance = pulse_duration * 17150
distance = round(distance, 2)

print(f"Distance: {distance} cm")
GPIO.cleanup()
```

✅ Camera Test:

```
bash
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libcamera-still -o test.jpg
```

✅ Coral USB Test:

```
bash
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lsusb
# Look for "Google Inc." — shows it's detected
```

💡 3.6 WIRING TIPS FOR BEGINNERS

- 🏷️ Label your wires with stickers or masking tape
- 🧪 Use a breadboard **first** for prototyping
- 🔌 Always power **off** before making or changing connections
- 🧹 Avoid overlapping or messy wire bundles
- 💧 Double-wrap all external connections with waterproof tape

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🎁 3.7 HOW TO PLACE COMPONENTS IN THE POUCH

Component	Position in Pouch
Raspberry Pi	Inside a plastic waterproof box
Coral USB	Plugged into Pi inside the box
Power Bank	Tightly placed with cable ties
Wires	Routed from stick → pouch using sleeve
Camera	Mounted outside front of pouch or chest strap

✅ 3.8 CHECKLIST BEFORE MOVING TO CODING

☐ Ultrasonic sensors respond with correct distance
☐ Camera is connected and captures image
☐ Coral USB is detected (`lsusb`)
☐ Speaker/earphones produce sound
☐ Power is stable, no shutdowns
☐ Wires are neatly routed and waterproofed

🧠 Final Words (Section 3 Summary)

You've now wired up and physically built the **entire system** that will power Saarthi. This is the **foundation** of your project.

Next up in **Section 4**, we'll dive into:

- Installing the OS and dependencies on Raspberry Pi
- Writing the main control code
- Running your first object detection model